

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: **udp port 53 unreachable**

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: **Evidence from the log entry for ICMP 203.0.113.2 marks the beginning of an error notification, signifying that the UDP data packet failed to reach its destination on DNS server port 53.**

The port noted in the error message is used for: **Port 53 is a port for DNS service**

The most likely issue is: **UDP message requesting an IP address for the domain "www.yummyrecipesforme.com" did not go through to the DNS server because no service was listening on the receiving DNS port**

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: **time is 1:24 p.m., 32.192571 seconds.**

Explain how the IT team became aware of the incident: **Numerous patrons reached out to report an inability to reach the organization's site, www.yummyrecipesforme.com, further noting that an error message indicating a destination port unreachable status appeared following an extended loading period.**

Explain the actions taken by the IT department to investigate the incident: **Initially, an attempt to access the site confirms the "destination port unreachable" error. To facilitate further investigation, the tcpdump network analysis tool was deployed during a subsequent page load attempt. The browser initiates a UDP-based DNS query to resolve the domain's IP address, a prerequisite for establishing an HTTPS connection. However, the traffic logs reveal that outgoing UDP packets are met with an ICMP response indicating udp port 53 unreachable.**

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

Note a likely cause of the incident:**Examination of the traffic logs confirms that ICMP echo reply messages were dispatched on two subsequent occasions, each resulting in the identical destination port unreachable notification.**